

A Pipeline for Effective and Extensible Stylization

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Figure 1: Traditional CG shot render (left), stylized image from automated render composite (middle), final stylized image from lighting (right). Our pipeline ensured that automated composites upstream from lighting would be representative of the final stylized result. © 2024 Disney

ABSTRACT

Inspired by turn-of-the-century watercolor illustrations, the art direction for Walt Disney Animation Studio’s “Wish” called for a unique watercolor storybook style. Traditional CG renders were insufficient to judge the final look of the film without additional processing in departments upstream from lighting. In order to prevent this issue, we created a stylization pipeline to (1) automatically provide all upstream departments stylized renders representative of the final stylized look of the film, (2) provide the ability to iterate closely with lighting to when stylization changes were needed, and (3) provide a centralized area for lighting to manage sweeping stylization changes across the show.

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1 INTRODUCTION

Stylized shows often depend on view-dependent effects that only make sense in camera space and after final lighting and compositing have been applied. As a result, while asset departments spend a lot of time building assets to fit the style of the show, stylized shows require a fair amount of shot-dependent fixes in compositing to achieve the final look of the film. Stylized shows face such challenges due to (1) the inability to visualize assets in asset development or character / environment performance in upstream

shot departments with the final stylization, and (2) the lack of frameworks to test and propagate stylization setups early in asset development and shot issuing.

The stylized art direction of “Wish” put a heavy emphasis on paper texture, linework, and watercolor effects such as ink pooling. On previous stylized projects at our studio, limited purpose-built toolsets were developed to alleviate some of these visualization issues. Learning from experiences on previous stylized shorts, we developed a configurable stylization pipeline that gave (1) all upstream departments imagery that was as representative of the show’s final style as possible, (2) all upstream departments the ability to iterate quickly with lighting when changes were needed, and (3) lighting the ability to manage show stylization and sequence / shot overrides in a centralized manner.

2 STYLIZATION PIPELINE

2.1 Automated Render Composite

The stylization pipeline is designed to empower both asset and shot artists to quickly iterate on stylization and to visualize the final stylized look. Thus, the system needs to be able to run arbitrary stylization treatments with arbitrary numbers of inputs and outputs on any offline render to accommodate changing treatments throughout development. Similarly, the system needs to be able to enable and disable required arbitrary output variables (AOVs) that usually are not used by upstream departments in all renders to accommodate any mattes / AOVs used by the stylization scripts.

On render, the automated render composite system turns on pre-configured AOVs that were specified by the selected stylization treatment. Upon main render completion in Disney’s Hyperion Renderer [Burley et al. 2018], the stylization treatment is configured to access the appropriate inputs and apply the treatment to produce a stylized render. The process is written to (1) trigger on render completion both in interactive sessions and in queue jobs, and (2) easily integrate into many other automated pipeline processes (e.g. set extension, stereo renders and look turntables).

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2.2 Stylization Management

Core stylistic elements general to all elements are driven by common sets of mattes and AOVs authored by lighting. For example, on "Wish", paper textures were applied to all elements on the show and were driven via shaders. Instead of individually adding these AOVs to every asset on the show, we provide a centralized way for lighting to edit mattes / AOVs / general materials in a single asset. We achieve this by introducing cross-asset USD composition arcs [Pixar 2016] that are added and configured on asset creation based on config-driven filter rules (e.g. environment, character, etc.).

2.3 Asset Stylization

To avoid unexpected stylization of assets in shots, look and lighting closely collaborated to create a system for look artists to author asset-specific mattes and AOVs to drive aspects of stylization. Lighting then uses these outputs to isolate and apply stylization as necessary. This workflow allows for look artists to paint detailed maps to control stylization, ranging from heaviness of paper texture and linework to numeric values on particular compositing operations. This workflow allows for asset variant specific setups that can be easily overridden. Using the automated render composite system, look artists are easily able to iterate and visualize their changes.

2.4 Iteration and Overrides

Our tooling gives artists the ability to apply the latest show-wide stylization treatment with local overrides to current interactive renders on local disk to test stylization changes. Our system also allows for artists to apply test stylization changes provided by another artist; these overrides can be applied for both local renders and on the renderfarm. The former workflow is useful for shot artists to locally iterate on stylization to meet art direction, whereas the latter workflow is useful for asset artists iterating with lighting artists for visual development and determining sequence overrides.

Artists use our publishing system to stage changes. The staged stylization treatment can then be widely accessed for additional testing. At the end of the week, if there are any staged changes, all staged changes are automatically deployed show-wide and a summary of the changes is sent to a collective notification channel to alert all stakeholders. The automated weekly deployment can be overridden to immediately release urgent fixes.

In sequences and shots where stylization tweaks are required, we have provided tooling for artists to publish stylization template overrides at the sequence and shot levels. For an easier workflow, we provided a simple tool in Nuke to take the current session's compositing script and publish to the specified sequence or shot.

2.5 Debugging

Unlike traditional CG renders, stylization adds an additional layer of indirection between authored data and resulting imagery. When building stylization treatments, lighting adds debugging layers that split the stylization into visually logical image layers that roughly correspond with mattes and AOVs controlled by look. For example, on "Wish", lighting provided debugging layers for different types of linework, watercolor edges, paper texture, etc. The image layers are included in the stylized color image produced by the automatic render compositing system, so upstream departments can easily

view them to determine where artistic features are coming from and direct notes to pertinent departments.

The automatic render compositing system writes out a copy of the stylization script used to create the stylized color image for debugging purposes. On the renderfarm, the debugging script is first configured and written to disk and then run in a separate job to make it easier to deploy and re-run stylization fixes *en masse*. If the debugging script is missing, tooling wrapping the internals of our system can be called to recreate the script.

3 IMPACT AND DISCUSSION

The stylization pipeline was deployed early in production and was heavily used by the look and lighting departments to quickly iterate on the visual style of the film. Using local overrides for stylization treatments, a lighting artist could make changes in composite and hand a new treatment to a look artist to test and render imagery. This iteration helped artists to quickly respond to and provide new stylized visuals for approvals.

The pipeline was also used throughout production (1) to produce stylized upstream renders for daily work and artistic approvals and (2) to drive stylization consistency and changes for the show. As a result, the stylization treatment and the base lighting comp template had early and frequent testing. Weaknesses in the stylization template were identified early in the show, which allowed the team to not only investigate and identify the issues, but also properly allocate resources towards solutions. When solutions required show-wide treatment changes, changes could be tested and released to production without negatively impacting departments. The ability to reliably visualize the final look of the film allowed artists to confidently make changes to their work. Sequence and shot template overrides were used in some shots where the base treatment did not work for particular elements.

The single lighting asset to control general stylization mattes and AOVs helped to simplify show-wide changes. While changes were not frequent, it prevented churn and allowed for lighting to introduce new widely used mattes quickly. The cross-asset composition method has since been used in other applications to simplify applying show-wide changes from a singular asset.

While not an initial goal, the stylization pipeline helped in rendering and complexity optimization efforts on the film. Using stylized shot renders, we were able to quickly identify assets in shots that could be further optimized due to the show's stylization. As a result, we were able confidently to push the level of detail and culling even farther than we would have otherwise.

Due to the success of the stylization pipeline on "Wish", this pipeline is being used on future shows currently in production. The extensibility of the pipeline has allowed other shows to apply vastly different styles to their imagery with a few configuration changes, which helped jumpstart their stylization investigations.

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